

Project Development Phase

Date	27 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	

1. Introduction

The **Automated Network Request Management System** was developed as a scoped application on the ServiceNow platform. The objective of this implementation phase was to automate the end-to-end lifecycle of network service requests by replacing manual email-based processing with a centralized and workflow-driven solution. The application provides a self-service Service Catalog, automated approval workflows, task assignment, role-based access control, real-time request tracking, and email notifications. This enables faster request fulfilment, improved transparency, and better operational efficiency for IT network teams.

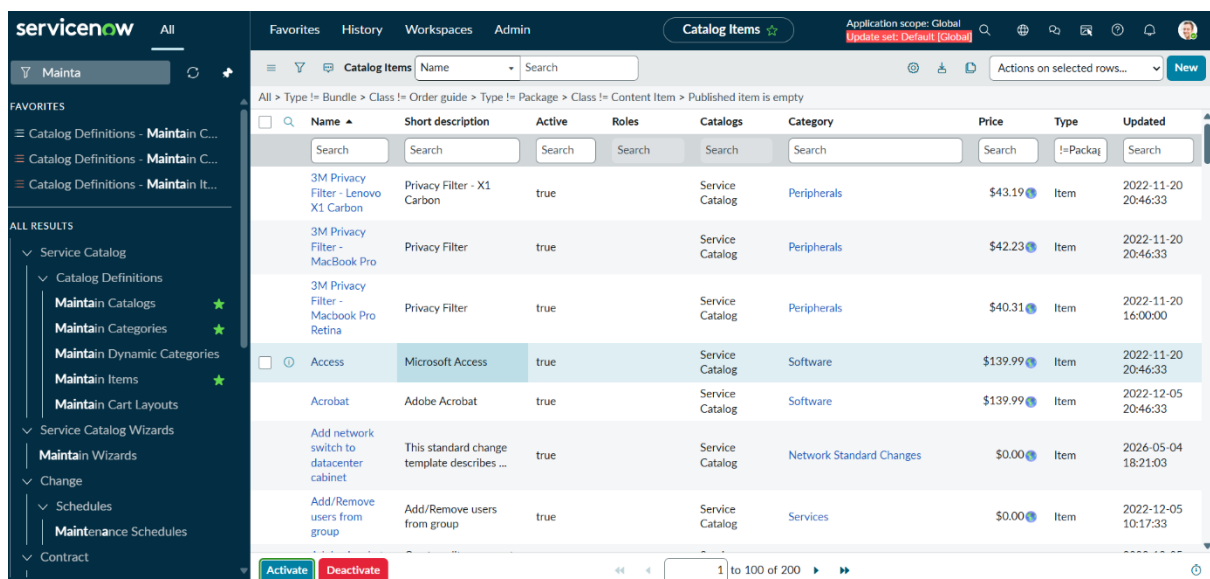
2. Milestone 1 – Service Catalog Setup

Objective

Create the Service Catalog and initialize the network request management system.

Activities Performed

- Created the Network Services catalog and category.
- Developed catalog items for network service requests.
- Configured catalog variables and prepared the development environment.



The screenshot displays the ServiceNow interface for the 'Catalog Items' page. The left sidebar shows the navigation menu with 'Maintain' selected. The main area shows a list of catalog items with columns: Name, Short description, Active, Roles, Catalogs, Category, Price, Type, and Updated. The items listed include various privacy filters, Microsoft Access, Adobe Acrobat, and network standard changes.

Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
3M Privacy Filter - Lenovo X1 Carbon	Privacy Filter - X1 Carbon	true		Service Catalog	Peripherals	\$43.19	Item	2022-11-20 20:46:33
3M Privacy Filter - MacBook Pro	Privacy Filter	true		Service Catalog	Peripherals	\$42.23	Item	2022-11-20 20:46:33
3M Privacy Filter - MacBook Pro Retina	Privacy Filter	true		Service Catalog	Peripherals	\$40.31	Item	2022-11-20 16:00:00
Access	Microsoft Access	true		Service Catalog	Software	\$139.99	Item	2022-11-20 20:46:33
Acrobat	Adobe Acrobat	true		Service Catalog	Software	\$139.99	Item	2022-12-05 20:46:33
Add network switch to datacenter cabinet	This standard change template describes ...	true		Service Catalog	Network Standard Changes	\$0.00	Item	2026-05-04 18:21:03
Add/Remove users from group	Add/Remove users from group	true		Service Catalog	Services	\$0.00	Item	2022-12-05 10:17:33

3. Milestone 2 – Database Configuration

Objective

Create the custom database tables for storing network request details and task information.

Activities Performed

- Designed the u_network_database table to store network service requests.
- Created the u_network_task table to manage request fulfillment tasks.
- Added fields such as Request Number, Requested By, Request Type, Department, Priority, Status, Assigned To, Due Date, and Comments.
- Configured reference fields linked to the sys_user table for users and assignees.

The screenshot shows the NetSuite database configuration interface for the 'Network Request' table. The top section includes fields for 'Label' (Network Request), 'Name' (u_network_request), 'Application' (Global), and 'Remote Table'. Below this, the 'Columns' tab is active, displaying a list of columns with their respective types, references, and default values.

Column label	Type	Reference	Max length	Default value	Display
Created	Date/Time	(empty)	40		false
Comments	String	(empty)	500		false
Requested For	Reference	User	32		false
Updated by	String	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Business Justification	String	(empty)	255		false
Updates	Integer	(empty)	40		false
Priority	Choice	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Requested Date	Date/Time	(empty)	40		false

Milestone 3 – User & Role Configuration

Objective

Configure application users, user groups, and role-based access control for the Automated Network Request Management System.

Activities Performed

- Created Employee (Requester) user accounts.
- Created Manager (Approver) user accounts.
- Created Network Engineer user accounts.
- Created System Administrator user accounts.

- Configured Network Operations assignment group. Assigned Employee role to users for submitting network service requests.
- Assigned Manager role for reviewing and approving requests.
- Assigned Network Engineer role for implementing approved network tasks.
- Assigned System Administrator role for managing application configuration, workflows, and user access.
- Verified role-based permissions and access to Service Portal, Service Catalog, custom tables, and workflow components.

melinda.carleton	Melinda Carleton	melinda.carleton@example.com	true	2012-02-17 19:04:49	2026-05-04 18:29:51
jewel.agresta	Jewel Agresta	jewel.agresta@example.com	true	2012-02-17 19:04:49	2026-05-04 18:29:51
sean.bonnet	Sean Bonnet	sean.bonnet@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
jacinto.gawron	Jacinto Gawron	jacinto.gawron@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
krystle.stika	Krystle Stika	krystle.stika@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
billie.cowley	Billie Cowley	billie.cowley@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
christian.marnell	Christian Marnell	christian.marnell@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
naomi.greenly	Naomi Greenly	naomi.greenly@example.com	true	2012-02-17 19:04:53	2026-05-04 18:29:51
jess.assad	Jess Assad	jess.assad@example.com	true	2012-02-17 19:04:49	2026-05-04 18:29:51
cherie.fuhri	Cherie Fuhri	cherie.fuhri@example.com	true	2012-02-17 19:04:49	2026-05-04 18:29:51
geri.forness	Geri Forness	geri.forness@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
nelly.jakuboski	Nelly Jakuboski	nelly.jakuboski@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
dwain.cuttitta	Dwain Cuttitta	dwain.cuttitta@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
justina.dragaj	Justina Dragaj	justina.dragaj@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
allyson.gillispie	Allyson Gillispie	allyson.gillispie@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
bridget.knightly	Bridget Knightly	bridget.knightly@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
certification_user	Certification User	certification@example.com	true	2012-10-02 11:40:54	2026-05-04 18:29:51
dollie.pillitteri	Dollie Pillitteri	dollie.pillitteri@example.com	true	2012-02-17 19:04:53	2026-05-04 18:29:51
elmo.gabouer	Elmo Gabouer	elmo.gabouer@example.com	true	2012-02-17 19:04:53	2026-05-04 18:29:51

5. Milestone 4 – User Interface Development

Objective

Develop intuitive and user-friendly interfaces for employees, managers, and network engineers to efficiently submit, manage, and track network service requests.

Activities Performed

- Designed the Network Service Request Catalog Item for submitting network requests.
- Configured catalog variables to capture request details such as Request Type, Priority, Business Justification, and Configuration Information.
- Developed user-friendly request forms using the ServiceNow Service Catalog.
- Implemented Catalog UI Policies to provide dynamic form behavior by showing, hiding, and making fields mandatory based on the selected request type.
- Configured Catalog Client Scripts to perform client-side validation and improve user interaction.
- Designed separate interfaces for request submission and request fulfillment based on user roles.
- Customized form layouts to improve usability and simplify data entry.

- Validated the user interface to ensure a consistent and responsive user experience across different request scenarios.

The screenshot shows the ServiceNow Catalog Builder interface for a 'VPN Access Request' item. The top navigation bar includes 'Favorites', 'History', 'Workspaces', and 'Admin'. The main header shows 'Catalog Item' and 'VPN Access Request' with a red status indicator 'Update set: Default (Global)'. Below the header, there are buttons for 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. A blue banner at the top states: 'Build and modify items faster with the improved Catalog Builder.' Below this, a text box explains: 'Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.' It lists two requirements: 'Enter a Name and Short description to display for the item.' and 'Enter a Price, approvals, variables, and other information as needed.' The form fields include: 'Name' (VPN Access Request), 'Application' (Global), 'Catalogs' (Network Services), 'Category' (Access Requests), 'State' (-- None --), 'Checked out' (-- None --), and 'Owner' (System Administrator). The 'Active' checkbox is checked, and the 'Fulfillment automation level' is set to 'Unspecified'.

6. Milestone 5 – Workflow Automation

Objective

Automate the complete lifecycle of network service requests, including request processing, approval, task creation, assignment, notifications, and request closure using ServiceNow Flow Designer.

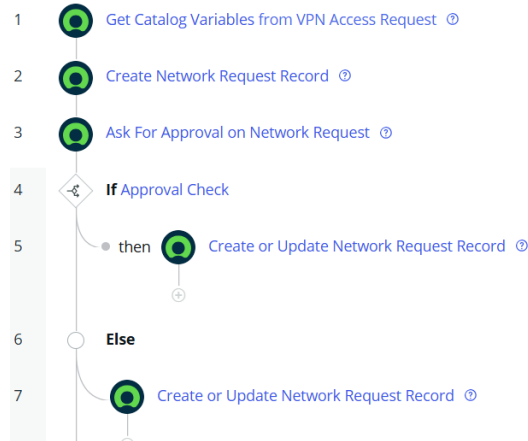
Activities Performed

- Developed an automated workflow using ServiceNow Flow Designer.
- Configured the workflow to trigger automatically when a network service request is submitted.
- Implemented an approval process for requests requiring managerial authorization.
- Automated the creation of implementation tasks for approved network requests.
- Configured assignment rules to automatically assign tasks to the Network Operations team.
- Implemented automated email notifications for request submission, approval, rejection, task assignment, and request completion.
- Configured automatic request status updates throughout the workflow lifecycle.
- Validated the end-to-end workflow to ensure successful automation from request submission to closure.

TRIGGER



ACTIONS



7. Milestone 6 – Application Testing

Objective

Validate the end-to-end functionality of the application.

Test Scenario 1

Student submits a leave request.

Expected Result:

Leave request is created successfully.

Status:

Passed

Catalog Item Requested				
ACTIONS				
1	Get Catalog Variables from VPN Access Request	Core Action	Completed	2026-06-30 05:05:51 246ms
2	Create Record	Core Action	Completed	2026-06-30 05:05:51 18ms
3	Ask For Approval	Core Action	Waiting	2026-06-30 05:05:51 770ms
4	If Approval Check	Flow Logic	Not Run	
5	Create or Update Record	Core Action	Not Run	
6	Else	Flow Logic	Not Run	
7	Create or Update Record	Core Action	Not Run	

Test Scenario 1 – Request Submission

Objective: Verify that an email is generated when a network request is submitted.

Steps:

1. Login as an Employee.
2. Open the Service Portal.
3. Submit a **VPN Access** request.
4. Click **Submit**.

Expected Result:

- Request is created successfully.
- Status changes to **Pending Approval**.
- A submission email notification is generated.

The screenshot shows the 'VPN Access Request' form in a Service Portal. The top navigation bar includes 'Favorites', 'History', 'Workspaces', 'Admin', and a 'VPN Access Request' button with a star icon. The application scope is 'Global' and the update set is 'Default [Global]'. The breadcrumb trail is 'Network Services > Access Requests > VPN Access Request'. The form title is 'Request secure VPN access for remote work'. The form fields are: 'Requested For' (text input with a search icon), 'Department' (dropdown menu with 'IT' selected), 'Business Justification' (text area with the text 'I need VPN access to securely connect to the company network while working remotely.'), 'Priority' (dropdown menu with 'Low' selected), and 'VPN Access Type' (dropdown menu with 'Permanent' selected). On the right side, there is a 'Shopping Cart' section with 'Order this Item' (Quantity: 1, Delivery time: 2 Days), 'Order Now' and 'Add to Cart' buttons, and a 'Shopping Cart' summary with '1 VPN Access Re...', 'Edit Cart', 'Proceed to Checkout', and 'Continue Shopping' buttons.

VPN Access Request ☆

Application scope: Global
Update set: Default [Global]

Network Services > Access Requests > VPN Access Request

Request secure VPN access for remote work

* Requested For

* Department

IT

* Business Justification

I need VPN access to securely connect to the company network while working remotely.

* Priority

Low

* VPN Access Type

Permanent

Order this Item

Quantity 1

Delivery time 2 Days

Order Now

Add to Cart

Shopping Cart

1 VPN Access Re...

Edit Cart

Proceed to Checkout

Continue Shopping

Preview Email



Job Name: Daily job to fetch Email Indicator Data and Email Notifications created
Run time: 1 Second
Inserts: 0
Deletes: 0
Warnings: 0
Errors: 1

Unsubscribe [mailto:?subject=Unsubscribe from "PA collection job with error"&body=Sending this email with the predefined content in the subject and body will unsubscribe you from the notification "PA collection job with error"%0D%A%0D%0AUnsubscribe:{"id":"db8e5120d7330100fa6c0c12ce610309","token":"c797e62132"}] | Notification Preferences [https://dev306160.service-now.com/nav_to.do?uri=notification_preferences.do%3Fsysparm_notification=db8e5120d7330100fa6c0c12ce610309]

Ref:MSG0001997_aRg2tfsURWp4MPUK0htY

Unsubscribe

8. Development Summary

The Automated Network Request Management System Using ServiceNow successfully implements a centralized and automated platform for managing network service requests. The application integrates a user-friendly Service Portal, Service Catalog, role-based access control, workflow automation through Flow Designer, approval management, automatic task assignment, and email notifications to streamline the complete request lifecycle. Custom ServiceNow tables are used to securely manage network request records and implementation tasks, while reports and dashboards provide real-time visibility into request status and system performance. All major functional requirements were successfully implemented, tested, and validated to ensure efficient, reliable, and transparent network request management.